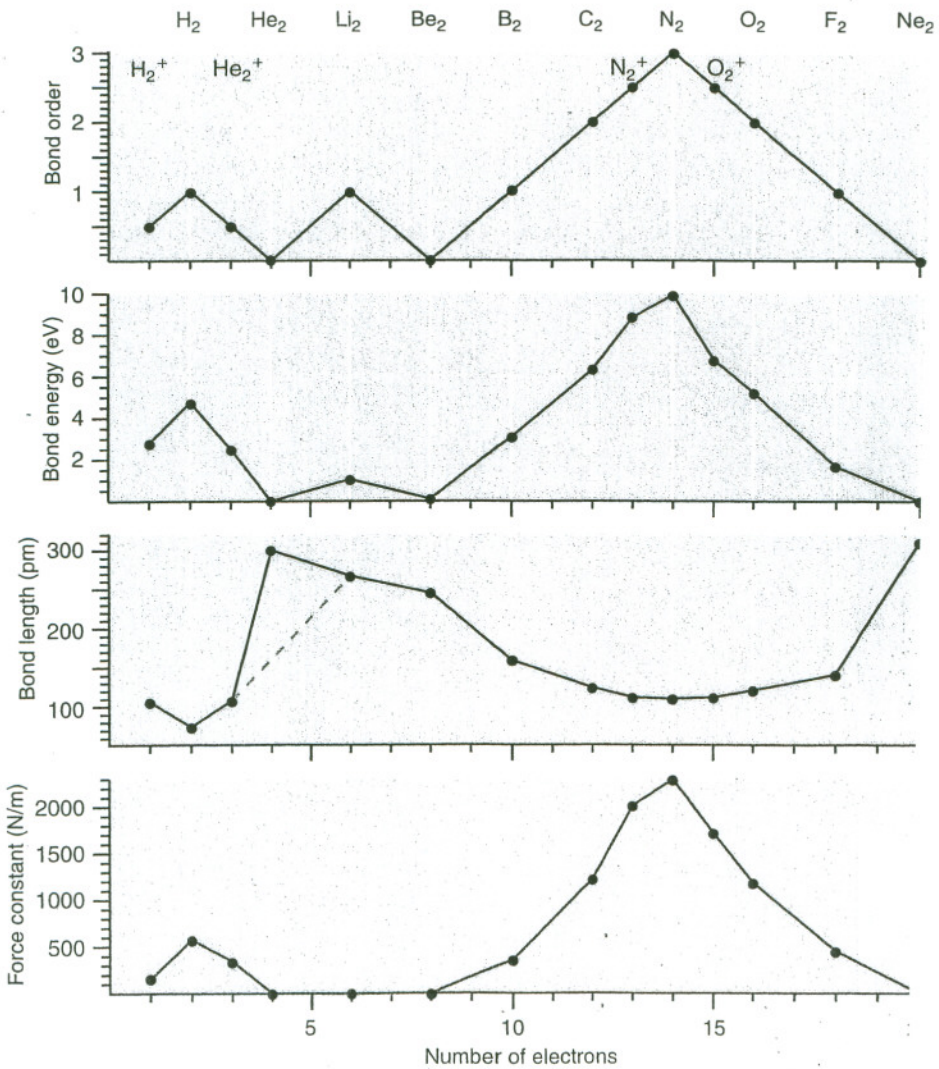
Physical Cume
February 9, 2007

Molecular Orbital Theory

Molecular orbital theory has shown its predictive power by providing an explanation for bond order, bond energy, bond length and vibrational force constant. Using the data shown below:

(1) Discuss how molecular orbital theory relates to bond order.

(2) Discuss the relationship between bond order and bond energy, bond length and vibrational force constant using molecular orbital theory.